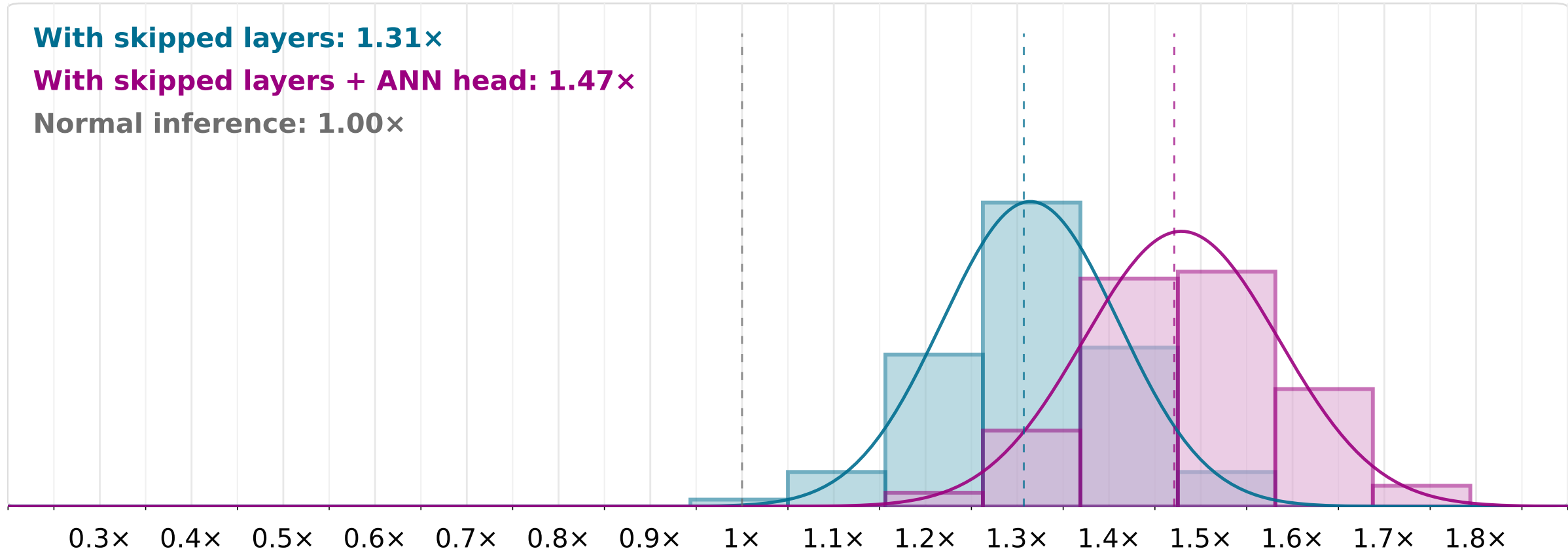


Self-speculation per-token speedup

Llama 3.2 3B Instruct



Setup

Prompt set = **python diverse**
Gap = **(1, 1)**
Block size = **2**
Layers = **28**
LM vocab = **128,256**
LM params = **3.21B**
Head params = **394.0M**
Head portion = **12.26%**

Runtime

GPU = **NVIDIA L4**
Backend = **sdpa**
Model dtype = **bfloat16**
Bridge dtype = **bfloat16**
Speed internals = **no**
Profile prompts = **15**
Warmup prompts = **5**
Measured prompts = **100**

Results

Peak mem normal = **6.06 GiB**
Mean speedup (w/o ANNH) = **1.31×**
Acceptance rate (w/o ANNH) = **44.7%**
Exact match (w/o ANNH) = **64.0%**
Peak mem self (w/o ANNH) = **6.06 GiB**
Mean speedup (w ANNH) = **1.47×**
Acceptance rate (w ANNH) = **44.6%**
Exact match (w ANNH) = **64.0%**
Peak mem self (w ANNH) = **6.12 GiB**
Tokens gen = **s 20,000 / n 20,000**

Profile

Drafter split (w/o ANNH) = **B 41.4% / H 52.6% / O 6.0%**
Verifier/normal (w/o ANNH) = **1.06×**
Drafter/normal (w/o ANNH) = **20.0%**
Drafter split (w ANNH) = **B 75.8% / H 13.7% / O 10.5%**
Verifier/normal (w ANNH) = **1.06×**
Drafter/normal (w ANNH) = **10.9%**
ANNH accuracy = **99.6%**
ANNH head speedup = **7.07×**
ANNH index = **8016 clusters; top-k 100**